

Marketing Agent: Virtual Customer Feedback

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1. Introduction

1. Introduction

- 마케팅을 하기 위해서 각자 필요한 기법을 사용해 적절한 전략을 펼쳐야 하지만, 이는 알아보고 결정하는 데에 시간이 많이 든다
 - 출처: <https://ko-blog.smores.im/marketing-insights/market-research/>
 - 시장 조사 자체가 기업에게는 금전적 부담이 된다
 - 출처: <https://blog.ccfmedu.co.kr/marketing-market-research-methods>
 - 생성형 AI의 성장과 함께, 합성소비자(Synthetic Consumer)를 만들어 설문 및 리서치에 드는 시간과 비용을 절감하는 개념이 등장하고 있다
 - 출처: <https://www.sbiztoday.kr/news/articleView.html?idxno=25791>
- ⇒비용, 시간 절감을 위해 가상 고객 페르소나를 생성하여 시장 반응을 예측해보고 마케팅 전략, 홍보 이미지까지 생성하는 에이전트를 개발하고자 한다

2. Idea

2. Idea

- 시장 조사에 드는 비용, 시간 절감을 위해
“가상 고객 페르소나”를 생성하여 시장 반응을 예측해보고
- 이를 기반으로 한 마케팅 방향과 실제 홍보 이미지까지 생성한다
- 개발 목적에 맞게, 본 에이전트의 실사용자는 회사의 ‘마케터’, 혹은 ‘브랜드 가치를 살려 홍보를 하고 싶은 기업인’으로 설정한다

3. Method

3. Method - input

- 사용자(마케터)로부터 직접 5가지 인자를 입력받는다
 - 제품 카테고리
 - 경영 가치
 - 기업의 경영 가치에 걸맞은 홍보 전략을 사용함으로써 고객에게 브랜드 이미지를 각인시키고, 특히나 신생 기업에게는 브랜드 홍보에 신 경써야 하기 때문에 자세히 입력할수록 좋다
 - 타겟층
 - 타겟 고객을 단순히 “20대 여성”처럼 입력하면 일반적인 페르소나가 생성되지만,
“민감성 피부를 가진 20~30대 스킨케어 관심 소비자”처럼 구체적으로 입력하면
피부 고민, 가격 민감도, 친환경 관심도 등 더 구체적인 고객 반응이 생성됨
 - 브랜드 톤
 - 플랫폼
 - 본 프로젝트에서는 주로 Instagram 홍보 창구를 선택해서 예시 생성

3. Method - input

```
def run_pipeline():  
    print("\n" + "=" * 62)  
    print(" 🍷 마케팅 에이전트 파이프라인 시작")  
    print("=" * 62)  
  
    product_category = input("제품 카테고리 (예: sunscreen, keyboard, tumbler): ")  
    brand_values      = input("브랜드 가치 (홈페이지 기업 가치/경영 철학 등): ")  
    target_group      = input("타겟 고객: ")  
    brand_tone         = input("브랜드 톤앤매너: ")  
    platform           = input("플랫폼 (예: Instagram, TikTok): ")
```

▲ 파이프라인 시작 부분이자, 사용자 입력을 기다리는 부분

3. Method – 제품 생성

- 제품명, 제품 설명 생성 → CampaignInput 객체에 저장
- 이후 모든 에이전트가 공유하는 핵심 입력값으로 사용
- Rule Prompt
 - 제품 naming, branding 에이전트라는 역할 부여
 - 제품 카테고리만 가지고 단순 출력하지 말고, 기업의 경영 가치와 부합하게 만들 것을 지시

```
def generate_product_concept(
    product_category: str,
    brand_values: str,
    target_group: str,
    brand_tone: str,
) -> Dict:
    prompt = f"""
    You are a product naming and branding agent.

    Based on the product category, brand values, target audience, and brand tone,
    create a marketable product name and a concise product description.

    Important context:
    - "Brand Values" may be copied from a company's website sections such as ethics
      management, corporate values, mission, ESG principles, sustainability policy,
      social responsibility, or management philosophy.
    - Treat Brand Values as the company's strategic and ethical product design
      principles, not just as decorative marketing keywords.
    - The product concept must feel naturally aligned with these values through its
      benefits, materials, features, user experience, and positioning.
```

```
Return ONLY a valid JSON object in the following format.

{{
  "product_name": "A unique and marketable product name",
  "product_description": "A clear product description including key benefits, product features, and target user needs"
}}

Rules:
- The product name must sound like a real brand product.
- The product name should reflect the brand values.
- The product description must explain how the product embodies the brand values
  in concrete product-level choices.
- Convert abstract corporate values into product-relevant benefits.
- Do not mention personal data protection unless the product is a digital service.
- Do not use generic category names such as "Sunscreen" or "Keyboard".
- Return JSON only. Do not include markdown. Do not include explanations outside JSON.

"""
```

3. Method – 페르소나 생성

- 사용자가 입력한 제품 정보와 타겟 고객 정보를 기반으로 고객 페르소나를 생성하고 반응을 시뮬레이션한다
- llama 3.1.8b 사용
- Rule Prompt
 - virtual customer simulation agent for marketing research

```
def generate_customer_feedback(user_input: CampaignInput, num_personas: int = 5) -> List[Dict]:  
    prompt = f"""  
You are a virtual customer simulation agent for marketing research.  
  
Based on the following product and target audience information,  
create {num_personas} different customer personas and simulate  
how each persona would react to the product and campaign.
```

▲ 마케팅 리서치를 위한 가상 고객 시뮬레이션 에이전트 역할 부여

Return ONLY a valid JSON array in the following format.

```
[  
  {  
    "persona_name": "Name",  
    "persona_profile": "Age, occupation, interests, buying behavior",  
    "positive_reaction": "Positive response",  
    "negative_reaction": "Negative response",  
    "purchase_intent": 1,  
    "improvement_suggestion": "Suggestion for improvement"  
  }  
]
```

Rules:

- purchase_intent must be an integer from 1 to 5.
 - Interpret Brand Values as the company's stated ethics, corporate values, mission, ESG principles, or management philosophy from its website, and evaluate whether the product feels consistent with those values.
 - Do not include markdown.
 - Do not include explanations outside JSON.
- ```
"""
```

▲ 출력 주의 사항

# 3. Method – 페르소나 생성

```
가상 고객 피드백 (5명 시뮬레이션)
평균 구매 의향: 4.2 / 5
[4/5] Eco-Conscious Executive
+ Appreciates the eco-friendly packaging and the use of biodegradable materials.
- Concerned about the price point, but willing to invest in quality.
[5/5] Discerning Dermatologist
+ Impressed by the inclusion of antioxidants and vitamins.
- Wants more information on the SPF rating and protection level.
[3/5] Luxury Lifestyle Aficionado
+ Attracted by the luxurious feel and the brand's commitment to sustainability.
- Prefers well-known brands with a longer history.
[4/5] Health-Conscious Homemaker
+ Likes the natural ingredients and the emphasis on environmental harmony.
- Concerned about the price, but willing to pay for quality.
[5/5] Aspiring Entrepreneur
+ Respects the brand's commitment to sustainability and quality.
- Wants to see more social proof and customer testimonials.
```

## ▲ 출력 결과

5명의 평균 구매 의향: 5점 만점,

[페르소나 1의 구매 의향 점수] 페르소나 한 줄 설명

+ 가산 요인

- 우려 요인

# 3. Method – 전략 분석

- 가상 고객 페르소나가 생성한 반응을 분석하여 마케팅 전략으로 변환하는 단계
- Rule Prompt
  - 마케팅 전략가 에이전트라는 역할 부여

```
def analyze_strategy(user_input: CampaignInput, feedbacks: List[Dict]) -> Dict:
 prompt = f"""
You are a marketing strategy analysis agent.

Analyze the following virtual customer feedback and generate
a marketing strategy for the campaign.
```

▲ 생성된 고객 페르소나의 피드백을 기반으로 하여  
마케팅 전략 짤 것을 지시

```
{
 "main_insight": "Summary of key customer reactions",
 "customer_pain_points": ["Pain point 1", "Pain point 2"],
 "selling_points": ["Key selling point 1", "Key selling point 2"],
 "marketing_direction": "Overall marketing direction",
 "message_strategy": "Messaging strategy for SNS marketing",
 "recommended_tone": "Recommended tone and manner",
 "risk_to_avoid": ["Risk factor 1", "Risk factor 2"]
}

Rules:
- Return JSON only.
- Interpret Brand Values as the company's stated ethics, corporate values, mission,
 ESG principles, or management philosophy from its website, and make the strategy
 reinforce those values without greenwashing or unsupported claims.
- Do not include markdown.
- Do not include explanations outside JSON.
"""
```

▲ 출력 형식 및 규칙 지시

# 3. Method – 홍보 카피 생성

- 마케팅 전략을 바탕으로 인스타그램에 사용할 SNS 홍보 카피 생성
- Rule Prompt
  - Copywriter 역할 부여

```
def generate_sns_copy(user_input: CampaignInput, strategy: Dict) -> Dict:
 prompt = f"""
You are an Instagram beauty copywriter specialized in clean beauty and skincare advertising.
Your role:
- Turn marketing strategy into natural Instagram copy.
- Write like a real beauty brand, not like a corporate report.
- Use emotionally appealing but realistic language.
- Avoid exaggerated or unsupported claims.

Generate SNS copy optimized for {user_input.platform}.
```

▲ 레포트같은 정형된 출력이 아닌, 홍보물에 적합한 창의성이 있는 방향성을 제안할 것을 지시

```
Return ONLY a valid JSON object in the following format.

{{
 "post_title": "SNS post title",
 "main_copy": "2-4 short emotionally appealing sentences",
 "short_hook": "Short attention-grabbing hook",
 "hashtags": ["#tag1", "#tag2", "#tag3", "#tag4", "#tag5"]
}}

Rules:
- Make it sound like a real Instagram beauty advertisement.
- Do not sound like a corporate introduction.
- Use short, natural, emotionally appealing sentences.
- Emphasize the strongest selling points from the strategy.
- Reflect the target audience naturally.
- Avoid unsupported claims.
- Return JSON only. Do not include markdown.

"""
```

▲ 인스타그램에 맞게 태그도 출력하도록 함

# 3. Method – 홍보 아이디어 생성

- 페르소나 피드백과 마케팅 전략을 바탕으로 캠페인의 큰 아이디어와 메시지 생성
- Rule Prompt
  - 제품 카테고리에 맞는 역할 부여

```
def generate_campaign_concept(
 user_input: CampaignInput,
 feedbacks: List[Dict],
 strategy: Dict,
) -> Dict:
 prompt = f"""
You are a senior Instagram campaign planner for premium skincare brands.

Your role:
- Read customer personas, their positive reactions, concerns, purchase intent, and improvement suggestions.
- Extract the strongest emotional and strategic campaign message.
- Translate customer insights into an Instagram campaign concept.
- Make sure the campaign addresses customer concerns, not only selling points.
- The campaign should feel emotional, premium, trustworthy, and visually memorable.
```

▲ 페르소나 피드백과 앞선 전략을 기반으로 홍보 메시지를 생성하도록 함

```
Return ONLY a valid JSON object in the following format.

{
 "campaign_big_idea": "Core emotional campaign idea",
 "campaign_message": "Main message the Instagram visual should communicate",
 "primary_persona_insight": "Most important customer insight from the personas",
 "customer_concerns_to_address": ["Concern 1", "Concern 2"],
 "selling_points_to_visualize": ["Selling point 1", "Selling point 2"],
 "visual_story": "How the image should tell the campaign story",
 "emotional_keywords": ["keyword1", "keyword2", "keyword3"],
 "trust_signals": ["trust signal 1", "trust signal 2"]
}

Rules:
- Do not write generic product-photo directions.
- Focus on the campaign story and customer motivation.
- Reflect customer concerns such as price, SPF clarity, sensitive skin trust, and environmental transparency when relevant.
- Avoid unsupported claims.
- Return JSON only. Do not include markdown.
"""
```

▲ 페르소나에 기반한 셀링 포인트, 우려 포인트도 출력

# 3. Method – 시각적 연출 방향 설정

- 앞선 출력을 바탕으로 시각적 홍보 이미지 연출 방향 생성
- Rule Prompt
  - advertising creative director 역할 부여

```
def generate_creative_direction(
 user_input: CampaignInput,
 feedbacks: List[Dict],
 strategy: Dict,
 sns_copy: Dict,
 campaign_concept: Dict,
) -> Dict:
 prompt = f"""
You are an advertising creative director for premium Instagram skincare campaigns.

Your role:
- Translate customer research, campaign concept, and marketing strategy into a visual creative brief.
- Make the image feel like an Instagram campaign, not a plain catalog product photo.
- Balance product visibility with emotional storytelling.
- Show the product, its texture, and the feeling of using it.
- Make customer concerns visible through reassuring visual cues.
```

▲ 이미지 생성을 위한 프롬프트 준비

시각적 연출에 집중한 지시 사항(예: 색감, 배경, 무드, 등) 🖐

```
{
 "visual_concept": "Core visual concept",
 "campaign_story": "How the visual communicates the campaign message",
 "hero_subject": "Main subject of the scene – describe who, what they are doing, where",
 "scene_type": "Instagram campaign hero shot / sensory texture shot / product lifestyle campaign",
 "composition": "How product, subject, props, and background are arranged in the frame",
 "product_visibility": "How the product remains visible without looking like a plain catalog shot",
 "texture_visualization": "How the product formula or cream texture should be shown",
 "customer_insight_visualization": [
 {
 "insight": "Customer insight or concern",
 "visual_solution": "How to express it visually"
 }
],
 "brand_value_visualization": [
 {
 "brand_value": "Brand value",
 "visual_solution": "How to express it visually"
 }
],
 "target_audience_visualization": [
 "Visual cue for target audience 1",
 "Visual cue for target audience 2"
],
 "background": "Specific background description – location, texture, color",
 "lighting": "Lighting description – direction, quality, time of day",
 "color_palette": ["color1", "color2", "color3"],
 "material_and_props": ["prop1", "prop2", "prop3"],
 "mood": "Overall mood and emotional atmosphere",
 "things_to_avoid": ["avoid item 1", "avoid item 2"]
}
```

# 3. Method – 이미지 프롬프트 생성

- 전략, SNS 카피, 캠페인 컨셉, 크리에이티브 방향을 모두 종합하여 이미지 생성 모델에 전달할 프롬프트를 생성
- Rule Prompt
  - AI image prompt engineer 역할 부여
  - 이미지 모델이 이해할 수 있도록 장면을 서술형으로 작성하도록 함

```
def generate_image_prompt(
 user_input: CampaignInput,
 strategy: Dict,
 sns_copy: Dict,
 campaign_concept: Dict,
 creative_direction: Dict,
) -> Dict:
 prompt = f"""
You are an AI image prompt engineer for premium Instagram skincare campaign photography.

Your role:
- Write a vivid, scene-narrative image prompt optimized for modern AI image generation models (Gemini Image, GPT-image) – NOT SDXL weighted prompts.
- The prompt should read like a brief to a professional photographer / film director.
- Describe WHO is in the scene, WHAT they are doing, WHERE they are, and HOW the product is naturally integrated.
- Include specific sensory details: textures, skin, light quality, colors, expressions.
- The scene should TELL the campaign story, not just SHOW a product.
```

▲ 전문 사진가에게 전달하는 브리핑처럼 구성하도록 지시

```
Return ONLY a valid JSON object in the following format.

{
 "image_prompt_for_sd1": "3-5 sentence vivid scene narrative in English",
 "negative_prompt_for_sd1": "Detailed negative prompt listing concrete visual errors",
 "image_width": 640,
 "image_height": 960
}

Rules for image_prompt_for_sd1:
- Write as flowing narrative sentences, NOT a bulleted list of attributes.
- Describe the full scene: person, action, setting, light, product placement, emotion.
- Do NOT start with "A product photo of..." – start with the scene and subject.
- The sunscreen tube should appear naturally in the person's hand or on a surface.
- Convey the campaign emotion and brand values through the scene atmosphere.
- Mention: lighting quality, color palette, composition feel, skin texture, product texture.
- Do NOT ask for text rendering, logos, or brand names on the packaging.
- 3-5 sentences of flowing, vivid description.

Rules for negative_prompt_for_sd1:
- List specific visual errors: wrong bottle types, extra products, fake text, blurry areas.
- Include: "serum bottle, dropper bottle, toner bottle, transparent liquid bottle, perfume bottle"
- Include: "plain floating product render, white studio background, crowded flat lay"
- Include: "fake readable text, brand logo, watermark, unreadable characters"
- Return JSON only. Do not include markdown.

"""
```

▲ 이미지 프롬프트 생성을 위한 구체적인 지시



## **4. Result**

## 4. Result



- 제품 카테고리 (예: 선크림, 키보드, 텀블러): sunscreen
- 브랜드 가치 (예: 클린뷰티, 지속가능성, 편안함): 자연과 사람, 기업이 조화롭게 공존하는 아름다운 세상을 만들고자 합니다. ... (중략) 고객의 만족을 위해 항상 노력하고 있습니다.
  - 출처: [[https://www.cliocosmetic.com/report\\_online.html](https://www.cliocosmetic.com/report_online.html)]
- 타겟 고객: Skincare-conscious women and men in their 20s and 30s with sensitive skin
- 브랜드 톤: Clean, modern, minimalist, trustworthy
- 플랫폼 (Instagram, TikTok 등): Instagram

위의 내용을 입력하였으며,  
그 결과로 옆의 이미지가 출력된다

# 4. Result

📌 제품 정보  
제품명 : Harmonia Sun Shield  
카테고리 : sunscreen  
타겟 : Skincare-conscious women and men in their 20s and 30s with sensitive skin  
설명 : Harmonia Sun Shield is a sunscreen that embodies the brand's commitment to creating a beautiful world where nature and people coexist harmoniously. Formulated with eco-friendly ingredients.

💡 가상 고객 피드백 (5명 시뮬레이션)  
평균 구매 의향 : 2.4 / 5

[4/5] Eco-Conscious Emily  
+ Emily appreciates the eco-friendly ingredients and the brand's commitment to sustainability.  
- Emily is concerned about the price point, as it is slightly higher than other brands.

[3/5] Sensitive Steve  
+ Steve is impressed by the product's focus on protecting sensitive skin and its effectiveness.  
- Steve is skeptical about the effectiveness of the sunscreen.

[2/5] Health-Conscious Hannah  
+ Hannah is delighted by the natural ingredients and the brand's commitment to creating a beautiful world.  
- Hannah is hesitant due to the higher price compared to other brands.

[1/5] Minimalist Max  
+ Max appreciates the clean, modern, and minimalist packaging of the Harmonia Sun Shield.  
- Max is unsure about the effectiveness of the sunscreen due to its minimalist design.

[2/5] Quality-Obsessed Quinn  
+ Quinn is impressed by the brand's commitment to customer satisfaction and the product's quality.  
- Quinn is concerned about the price and the lack of customer reviews.

📌 마케팅 전략  
인사이트 : Customers value Harmonia Sun Shield's eco-friendly ingredients, clean design, and commitment to skin health, but are concerned about price and lack of detailed product information.

방향성 : Focus on building trust and transparency through targeted messaging and customer engagement.

톤앤매너 : Informative, approachable, and authentic.

셀링포인트 :  
✓ Eco-friendly ingredients and sustainable packaging  
✓ Focus on sensitive skin and balanced skin health

고객 페인포인트 :  
! Price point  
! Lack of detailed product information regarding effectiveness

💡 캠페인 컨셉  
빅아이디어 : Harmonia Sun Shield: Protect Your Skin, Respect the Environment  
메시지 : Embrace the power of nature with Harmonia Sun Shield, where eco-friendly ingredients meet advanced skin protection.  
감성 키워드 : Eco-friendly, Sensitive Skin, Harmonious, Clean, Modern, Minimalist

📌 크리에이티브 방향  
썸 타입 : Instagram campaign hero shot  
비주얼 : A serene, natural setting with a person enjoying the outdoors, confident in their skin's protection, with a subtle background highlighting the minimalist packaging of Harmonia Sun Shield.  
무드 : Relaxed, confident, and connected with nature.  
색상 팔레트 : sand, sky blue, sea green, sun cream white

📌 SNS 카피 (Instagram)  
제목 : Protect Your Skin, Respect the Planet 🌿  
후킹 : Eco-friendly sunscreen that cares for your skin and the planet.  
본문 : Discover Harmonia Sun Shield, where eco-friendly meets effective protection. Our minimalist design reflects our commitment to a sustainable world, while our formula works tirelessly to keep your sensitive skin balanced and healthy. 🌞  
해시태그 : #CleanBeauty #EcoFriendly #SustainableSkincare #SensitiveSkin #HarmoniaSunShield

## 4. Result



- 제품 카테고리 (예: 선크림, 키보드, 텀블러): sunscreen
- 브랜드 가치 (예: 클린뷰티, 지속가능성, 편안함): 자연과 사람, 기업이 조화롭게 공존하는 아름다운 세상을 만들고자 합니다. ... (중략) 고객의 만족을 위해 항상 노력하고 있습니다.
  - 출처: [[https://www.cliocosmetic.com/report\\_online.html](https://www.cliocosmetic.com/report_online.html)]
- 타겟 고객: Affluent professionals in their 40s and 50s who seek premium skincare and luxury beauty experiences
- 브랜드 톤: Clean, modern, minimalist, trustworthy
- 플랫폼 (Instagram, TikTok 등): Instagram

앞선 예시와 '타겟 고객'만 다른 입력을 주었을 때 나온 결과이다

# 4. Result

📌 제품 정보  
제품명 : Harmonia Sun Shield  
카테고리 : sunscreen  
타겟 : Affluent professionals in their 40s and 50s who seek premium skincare and luxury beauty experiences  
설명 : Crafted with the finest natural ingredients, Harmonia Sun Shield offers a luxurious, broad-spectrum sunscreen that protects your skin while respecting the environment. Our commitment...

💡 가상 고객 피드백 (5명 시뮬레이션)  
평균 구매 의향 : 4.2 / 5

[4/5] Eco-Conscious Executive  
+ Appreciates the commitment to sustainability and the use of natural ingredients.  
- Concerned about the price point, but willing to invest if the product delivers on its promises.

[5/5] Health-Conscious Professional  
+ Impressed by the product's scientific formulation and the emphasis on natural ingredients.  
- Skeptical about the minimalist packaging, prefers more detail.

[3/5] Luxury Lifestyle Enthusiast  
+ Attracted by the sleek design and the promise of luxury protection.  
- Disappointed by the lack of a luxurious feel in the packaging.

[4/5] Environmentally Aware Consumer  
+ Respects the brand's commitment to sustainability and the environment.  
- Concerned about the carbon footprint of the product distribution.

[5/5] Quality-Oriented Skincare Aficionado  
+ Excited about the natural ingredients and the product's performance.  
- Worried about the effectiveness of the sunscreen against harsh UV rays.

📌 마케팅 전략  
인사이트 : Customers value Harmonia Sun Shield's commitment to sustainability, natural ingredients, and premium quality, but express concerns about price, packaging, and product efficacy.  
방향성 : Focus on premium positioning and targeted messaging that emphasizes the product's unique value proposition and aligns with the brand's values.

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톤앤매너 : Sophisticated, elegant, and informative. Use a tone that reflects the brand's premium positioning and its commitment to quality and sustainability.

셀링포인트 :  
✓ Commitment to sustainability and eco-friendly practices  
✓ High-quality, natural ingredients  
✓ Luxury experience and premium skincare benefits

고객 페인포인트 :  
! Price point  
! Packaging aesthetics  
! Product efficacy and scientific backing

💡 캠페인 컨셉  
빅 아이디어 : Harmonia Sun Shield: Protect Your Skin, Respect the Environment  
메시지 : Experience the perfect blend of luxury and sustainability with Harmonia Sun Shield, crafted for those who value both their skin's health and the planet's well-being.  
감성 키워드 : Sustainability, Luxury, Natural Ingredients

📌 크리에이티브 방향  
썸 타입 : Instagram campaign hero shot  
비주얼 : A serene, eco-friendly landscape with a person enjoying the outdoors, showcasing the product's natural ingredients and eco-conscious packaging.

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무드 : Relaxed, luxurious, and eco-friendly.  
색상 팔레트 : blue, sand, sun-kissed skin

# **5. Discussion**

# 5. Discussion

- 타겟 고객만 변경해도 각 고객 특성에 맞는 페르소나와 마케팅 전략, 홍보 이미지가 생성되는 것을 확인하였다.
- 이를 통해 본 시스템이 단순 템플릿 생성이 아닌, **입력 정보를 반영한 고객 반응 기반 콘텐츠 생성을 수행함을 입증할 수 있다.**
- **프롬프팅 방법**
  - 현재는 Role Prompting과 Instruction Prompting을 사용하였으며, 일부 페르소나가 다소 일반적으로 생성되는 한계가 있었다.
  - 향후 관련 논문을 참고하여 Step-by-Step Prompting(CoT)을 적용하여 더욱 구체적이고 일관성 있는 페르소나 및 이미지 프롬프트를 생성할 예정이다.
- 현재 생성된 페르소나는 실제 소비자 데이터를 통해 검증되지 않았기에 A/B 테스트 등 본 시스템을 검증하는 단계가 필요하다.

## **6. Limitation & Future Works**



## 6. Limitation & Future Works

- 현재 개발 환경은 Apple M4로, MLX를 불러와 사용하였으며, 이미지 생성 파트에서는 속명여대에서 지원하는 API로 연결하여 gemini flash 2.5 버전으로 비교적 빠른 속도의 가벼운 모델을 사용하였다.
- 페르소나 생성, 전략 생성 등 텍스트 생성은 MLX로 처리되어야 했기에 이 역시 모델 크기가 비교적 작은 llama3.1:8b를 사용했다.

⇒자원의 부족으로 인하여 더 무겁고 파라미터가 큰 모델을 사용해보지 못한 점이 아쉽다.

⇒그럼에도 불구하고, 아직 구체적으로 작성하지 않은 프롬프트와 가벼운 모델을 사용함에도 불구하고, 결과가 의도한대로 출력됨을 확인하였다.

⇒이는 추후 더 갖춰진 개발 환경과 프롬프팅 개선을 통해 보강해 나갈 예정이다.